Conditional distribution of the time-average of a geometric Brownian motion and option pricing in the SABR model

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Abstract

An analytical approximation for the distribution of the time-average of a geometric Brownian motion conditional on its terminal value is proposed, following from an asymptotic expansion for the Hartman-Watson distribution. An error bound is available for the approximation error. We study the performance of this approximation and show good agreement with exact results for the density and moments of this distribution, for sufficiently small maturity. As an application we discuss option pricing in the correlated log-normal SABR model, and demonstrate good numerical performance for cases of practical interest.

Keywords: Asymptotic expansions, Option pricing, Stochastic processes

1. Introduction

The joint distribution of the time-integral of a geometric Brownian motion and of its terminal value appears in many problems of financial engineering, such as option pricing and simulation of the SABR model [14, 1], and the pricing of Asian-type options in the Black-Scholes model.

Define the standardized time-integral of the geometric Brownian motion

$$A_t^{(\mu)} := \int_0^t e^{2(B_s+\mu s)} ds \tag{1}$$

where $\{B_s\}_{s\geq 0}$ is a standard Brownian motion and denote $s_t^{(\mu)} := e^{B_t+\mu t}$.

The object of interest in this paper is the conditional distribution of $a_t^{(\mu)} := \frac{1}{t} A_t^{(\mu)}$ at given $s_t^{(\mu)} = s$. We will call this simply the *conditional distribution* of $a_t^{(\mu)}$, and denote it as $f_{a_t|s_t}(a, s, t)$.

Literature review. An exact result for the joint distribution of $A_t^{(\mu)}$ and s_t has been given by Yor [41], in terms of the Hartman-Watson integral $\theta(r,t)$. The precise result is given below in Section 2. The evaluation of this integral by numerical integration is challenging for small t [3], and requires the use of high precision numerics in intermediate steps. This motivated the search for alternative computation methods. We summarize briefly the numerical and analytical evaluation methods proposed in the literature.

Among the numerical methods we mention numerical evaluation by inversion of a Laplace transform of $\theta(r,t)$ with respect to t by Bernhart, Mai (2015) [5]. Cai, Song and Chen [7] proposed to evaluate θ by numerical inversion of its Laplace transform with respect to r. Antonov et al [1] propose a method for numerical evaluation by integration along a judiciously chosen contour in the complex plane. A different method using a contour integration method leading to analytical expressions was discussed in Lyasoff (2016) [25].

Moment matching methods are popular with practitioners due to their simplicity of use [19, 20, 27, 8]. These methods do not rely on the exact Yor result [41] for the joint distribution, but assume a simple functional form for this distribution, which is then constrained by imposing moments conditions satisfied by the exact distribution. An alternative approximation for the joint density has been proposed by Leitao Rodriguez, Grzelak and Oosterlee (2017) [36] using a copula method.

Several analytical approximations have been proposed, based on asymptotic expansions of $\theta(r,t)$ for small t. The leading asymptotics as $t \to 0$ at fixed r was obtained in [3] and refined in [12]. The $t \to 0$ asymptotics of $\theta(r,t)$ at fixed $\rho = rt$ was obtained in [32]. The latter asymptotics is relevant for the joint distribution of $(a_t^{(\mu)}, s_t^{(\mu)})$ as $t \to 0$, which will be seen to appear in practical applications, while the former is connected to the joint distribution of $(A_t^{(\mu)}, s_t^{(\mu)})$ in the same limit.

In this paper we will study the use of the asymptotic expansion of [32] as an analytical approximation for the conditional distribution $f_{a_t|s_t}(a,s,t)$. This approximation is given below in (18) and has the advantage that it is asymptotic to the exact distribution as $t \to 0$, see Proposition 1.

Some definitions. The conditional distribution is given by

$$f_{a_t|s_t}(a,s,t) = \frac{f(a,s,t)}{f_{\mathrm{LN}}(s,t;\mu)} \tag{2}$$

where $f(a,s,t)$ is the joint distribution of the two variables $(\frac{1}{t}A_t^{(\mu)}, s_t)$, defined

as

$$\mathbb{P}(a_t \in da, s_t \in ds) = f(a,s,t)\frac{dads}{a} \tag{3}$$

and

$$f_{\rm LN}(s,t;\mu) = \frac{1}{\sqrt{2\pi t}s} e^{-\frac{1}{2t}(\log s - \mu t)^2} \tag{4}$$

is the log-normal density of $s_t^{(\mu)}$.

The conditional distribution $f_{a_t|s_t}(a,s,t)$ is normalized to 1 as

$$\int_0^\infty f_{a_t|s_t}(a,s,t)\frac{da}{a} = 1 \tag{5}$$

such that a log-normal distribution is recovered for the marginal distribution over s

$$\int_0^\infty f(a,s,t)\frac{da}{a} = f_{\rm LN}(s,t;\mu)\,. \tag{6}$$

We present next an exact result for the conditional distribution which can be extracted from Yor [41], expressed as a one-dimensional integral. In Section 3 we present an analytical approximation which approaches asymptotically the exact result in the small-maturity limit $t \to 0$.

2. Exact result for the conditional distribution

An exact representation for the joint distribution $f(a,s,t)$ has been given by Yor [41]. In this paper the following result has been proven

$$\begin{aligned} &\mathbb{P}(A_t^{(\mu)} \in du, B_t + \mu t \in dx) = \\ &\quad e^{\mu x - \frac{1}{2}\mu^2 t} \exp\left(-\frac{1+e^{2x}}{2u}\right) \theta(e^x/u, t)\frac{dudx}{u}\,, \end{aligned} \tag{7}$$

where $\theta(r,t)$ is the Hartman-Watson integral, given by

$$\theta(r,t) = \frac{r}{\sqrt{2\pi^3 t}} e^{\frac{\pi^2}{2t}} \int_0^\infty e^{-\frac{1}{2t}\xi^2 - r\cosh\xi} \sinh\xi \sin\frac{\pi\xi}{t} d\xi\,. \tag{8}$$

Changing variables to $a_t^{(\mu)}$ and $s_t^{(\mu)}$ this gives

$$f(a,s,t) = s^{\mu-1} e^{-\frac{1}{2}\mu^2 t} e^{-\frac{1+s^2}{2at}} \theta\Big(\frac{s}{at}, t\Big)\,. \tag{9}$$

Substituting this joint distribution into (2) gives an exact result for the conditional distribution, expressed as a one-dimensional integral (the Hartman-Watson integral)

$$\begin{aligned} f_{a_t|s_t}(a,s,t) &= \sqrt{2\pi t}s^{\mu}e^{-\frac{1}{2}\mu^2 t}e^{-\frac{1+s^2}{2at}+\frac{1}{2t}(\log s-\mu t)^2}\theta\Big(\frac{s}{at},t\Big) \\ &= \sqrt{2\pi t}e^{-\frac{1}{t}(\frac{1+s^2}{2a}-\frac{1}{2}\log^2 s)}\theta\Big(\frac{s}{at},t\Big)\,. \end{aligned} \tag{10}$$

Remarkably, all dependence on μ has disappeared on the right-hand side. This is due to the cancellation of μ dependence in the ratio

$$\frac{1}{f_{LN}(s,t;\mu)}s^{\mu-1}e^{-\frac{1}{2}\mu^2 t} = \sqrt{2\pi t}s^{\mu}e^{-\frac{1}{2}\mu^2 t}e^{\frac{1}{2t}(\log s-\mu t)^2} = \sqrt{2\pi t}e^{\frac{1}{2t}\log^2 s}\,. \tag{11}$$

The numerical evaluation of the $\theta(r,t)$ integral in the small-t region is unstable unless very high precision is used. A possible alternative method is the use of analytic approximations in this region [3, 12, 32]. In Section 3 we give such an analytical approximation obtained by a small-t asymptotic expansion.

2.1. Moments of the conditional distribution

The first two conditional moments of a_t at given s_t are known exactly from [19, 27]. They are given in Proposition 1 of [19]. Translated to our notations, the first moment is

$$\mathbb{E}[a_t|s_t=s] = \frac{1}{2\sqrt{t}}\frac{\Phi(\frac{1}{\sqrt{t}}\log s+\sqrt{t})-\Phi(\frac{1}{\sqrt{t}}\log s-\sqrt{t})}{\phi(\frac{1}{\sqrt{t}}\log s+\sqrt{t})} \tag{12}$$

with $\phi(x)=\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}x^2}$ and $\Phi(x)=\int_{-\infty}^{x}\phi(t)dt$.

The second moment is

$$\begin{aligned} \mathbb{E}[(a_t)^2|s_t=s] = &-\frac{1}{4t^{\frac{3}{2}}}(1+s^2)\frac{\Phi(\frac{1}{\sqrt{t}}\log s+\sqrt{t})-\Phi(\frac{1}{\sqrt{t}}\log s-\sqrt{t})}{\phi(\frac{1}{\sqrt{t}}\log s+\sqrt{t})} \\ &+\frac{1}{4t^{\frac{3}{2}}}\frac{\Phi(\frac{1}{\sqrt{t}}\log s+2\sqrt{t})-\Phi(\frac{1}{\sqrt{t}}\log s-2\sqrt{t})}{\phi(\frac{1}{\sqrt{t}}\log s+2\sqrt{t})}\,. \end{aligned} \tag{13}$$

These moments have been used in the literature [19, 27] to construct a log-normal analytical approximation for the joint distribution with parameters chosen such that the first two moments are correctly reproduced.

3. Small-maturity asymptotics of the conditional distribution

An asymptotic expansion for the Hartman-Watson function $\theta(r,t)$ in the limit $t \to 0$ at fixed $rt = \rho$ has been given in [32]. The result is given below.

$$\theta(\rho/t, t) = \frac{1}{2\pi t} e^{-\frac{1}{t}(F(\rho)-\frac{\pi^2}{2})} \Big(G(\rho) + G_1(\rho)t + O(t^2) \Big) , \quad (t \to 0) \tag{14}$$

where $F(\rho), G(\rho), G_1(\rho), \cdots$ are known functions. They are given for convenience in Appendix A.

Denote the leading term in this expansion

$$\hat{\theta}(\rho/t, t) := \frac{1}{2\pi t} e^{-\frac{1}{t}(F(\rho)-\frac{\pi^2}{2})} G(\rho) \tag{15}$$

and write

$$\theta(\rho/t, t) = \hat{\theta}(\rho/t, t)(1 + \vartheta(\rho, t)) \tag{16}$$

with $\vartheta(\rho, t)$ an error term. Using numerical estimates of the integrand, an error bound was obtained in [33]. The error term $\vartheta(\rho, t)$ is bounded uniformly over ρ as

$$|\vartheta(\rho, t)| \leq \frac{1}{70} t \, . \tag{17}$$

Using the approximation (15) we obtain the following analytical approximation for the conditional distribution $f_{a_t|s_t}(a, s, t)$, which furthermore approaches asymptotically the exact result in the $t \to 0$ limit. We denote $f(a, s, t) \sim g(a, s, t)$ if $\lim_{t\to 0} f/g = 1$ for all $a, s > 0$.

Proposition 1. *The conditional distribution approaches the asymptotic result $f_{a_t|s_t}(a, s, t) \sim \hat{f}_{a_t|s_t}(a, s, t)$ as $t \to 0$ with*

$$\hat{f}_{a_t|s_t}(a, s, t) := \frac{1}{\sqrt{2\pi t}} e^{-\frac{1}{t}(H(a,s)-\frac{1}{2}\log^2 s)} G(s/a) \, , \tag{18}$$

with

$$H(a, s) := \frac{1 + s^2}{2a} + F(s/a) - \frac{\pi^2}{2} \, . \tag{19}$$

Proof. The proof follows by substituting the leading order term in the asymptotic expansion of the Hartman-Watson integral (14) into the representation (9) of the joint distribution $f(a, s, t)$, and using it to compute the conditional distribution. □

Remark 1. *Writing the approximation error of the asymptotic result as*

$$f_{a_t|s_t}(a,s;t) = \hat{f}_{a_t|s_t}(a,s;t)(1+\varepsilon(a,s,t))\,, \tag{20}$$

numerical estimates for the integrand yield the error bound [33] $|\varepsilon(a,s,t)| \leq \frac{1}{70}t$*, uniformly over* (a,s)*.*

We study next some properties of the asymptotic approximation for the conditional distribution $\hat{f}_{a_t|s_t}(a,s,t)$ given in (18).

4. Laplace transform of the Hartman-Watson distribution

One important property of the Hartman-Watson integral $\theta(r,t)$ is its Laplace transform with respect to r. The following result appears as Proposition 4.2 in Matsumoto and Yor [26]:

$$F_\theta(x,t) := \int_0^\infty e^{-rx}\theta(r,t)\frac{dr}{r} = \begin{cases} \frac{1}{\sqrt{2\pi t}}e^{-\frac{1}{2t}\arg\cosh^2 x} & \text{for } x \geq 1 \\ \frac{1}{\sqrt{2\pi t}}e^{+\frac{1}{2t}\arg\cos^2 x} & \text{for } |x| \leq 1 \end{cases} \tag{21}$$

Using this result one can prove the following exact result for the Laplace transform of $1/A_t^{(\mu)}$, conditional on s_t.

Proposition 2. *For any* $\mu \in \mathbb{R}$ *and* $\lambda \geq 0$*, we have*

$$\mathbb{E}\Big[e^{-\frac{\lambda}{A_t^{(\mu)}}}|s_t = s\Big] = \exp\Big(-\frac{1}{2t}\varphi_{\log s}(\lambda)\Big) \tag{22}$$

where

$$\varphi_x(\lambda) := \arg\cosh^2(\lambda e^{-x} + \cosh x) - x^2 \tag{23}$$

This appears as Theorem 5.6 in Matsumoto, Yor (2005) [26] for the Laplace transform of $1/A_t^{(\mu)}$.

Proof. The expectation (22) can be written as an integral over the conditional distribution

$$\begin{aligned} \mathbb{E}\Big[e^{-\frac{\lambda}{A_t^{(\mu)}}}|s_t = s\Big] &= \int_0^\infty e^{-\frac{\lambda}{at}} f_{a_t|s_t}(a,s,t)\frac{da}{a} \\ &= \frac{1}{f_{LN}(s,t)} s^{\mu-1} e^{-\frac{1}{2}\mu^2 t} \int_0^\infty e^{-\frac{\lambda}{at}-\frac{1+s^2}{2at}}\theta(\frac{s}{at},t)\frac{da}{a}\,. \end{aligned} \tag{24}$$

Introduce a new integration variable $r = \frac{s}{at}$. The integral is evaluated as

$$\int_0^\infty e^{-\frac{\lambda}{at}-\frac{1+s^2}{2at}}\theta(\frac{s}{at},t)\frac{da}{a} = \int_0^\infty e^{-r(\frac{\lambda}{s}+\frac{1+s^2}{2s})}\theta(r,t)\frac{dr}{r} = F_\theta(x_*,t) \tag{25}$$

with $x_* := \frac{\lambda}{s} + \frac{1+s^2}{2s}$ and $F_\theta(x,t)$ is defined in (21). Since $\frac{1+s^2}{2s} \geq 1$ for all $s \geq 0$, and $\lambda > 0$ by assumption, we can use the upper branch of $F_\theta(x,t)$ in (21), with

$$\begin{aligned} F_\theta(x_*,t) &= \frac{1}{\sqrt{2\pi t}}\exp\Big(-\frac{1}{2t}\arg\cosh^2\Big(\frac{\lambda}{s}+\frac{1+s^2}{2s})\Big) \\ &= \frac{1}{\sqrt{2\pi t}}\exp\Big(-\frac{1}{2t}\arg\cosh^2(\lambda e^{-\log s}+\cosh\log s)\Big)\,. \end{aligned} \tag{26}$$

Substituting the expression (26) into (24) yields the stated result (22).

□

Short maturity limit for the conditional distribution. Denote the random variable $a_t(s) := (a_t|s_t = s)$. We prove here a result for the short maturity limit of $a_t(s)$.

Proposition 3. *As $t \to 0$ we have the convergence in probability*

$$\lim_{t\to 0} a_t(s) \stackrel{p}{=} \frac{s^2-1}{2\log s}\,. \tag{27}$$

Proof. It is sufficient to prove convergence in distribution to a constant, which implies convergence in probability.

The moment-generating function of $1/a_t^{(\mu)}$ is obtained from (22) as

$$\mathbb{E}\Big[e^{-\frac{\lambda}{a_t^{(\mu)}}}|s_t = s\Big] = \exp\Big(-\frac{1}{2t}\varphi_{\log s}(\lambda t)\Big) \tag{28}$$

Expanding $\varphi_x(\lambda)$ to linear order in λ we have

$$\varphi_x(\lambda) = \lambda\frac{2xe^{-x}}{\sinh x} + O(\lambda^2) \tag{29}$$

Substituting into (28) and taking the $t \to 0$ limit gives

$$\lim_{t\to 0}\mathbb{E}\Big[e^{-\frac{\lambda}{a_t^{(\mu)}}}|s_t = s\Big] = \exp\Big(-\lambda\frac{2\log s}{s^2-1}\Big)\,. \tag{30}$$

By the continuity theorem, pointwise convergence of the moment generating function implies weak convergence for $a_t(s)$ as $t \to 0$, see for example Chapter XIII.1 in [10], vol. 2. The expression on the right hand side is the moment generating function of a constant value, which gives the stated result (27).

□

4.1. Small-maturity inversion of the Laplace transform

We give here an asymptotic inversion of the Laplace transform $F_\theta(x,t)$ for $x \geq 1$ in the small-maturity limit $t \to 0$.

Define the approximation for the Hartman-Watson integral obtained by keeping only the leading order term in the asymptotic expansion

$$\hat{\theta}(r,t) := \frac{1}{2\pi t} G(rt) e^{-\frac{1}{t}(F(rt)-\frac{\pi^2}{2})} \tag{31}$$

It is natural to ask how close does the Laplace transform of this function (in the r variable) approximates the exact result (21). We prove here that this Laplace transform is asymptotic to the exact result as $t \to 0$.

Proposition 4. *Define*

$$\hat{F}_\theta(x,t) := \int_0^\infty e^{-rx} \hat{\theta}(r,t) \frac{dr}{r} \,. \tag{32}$$

For any $x \geq 1$ we have

$$\hat{F}_\theta(x,t) \sim \frac{1}{\sqrt{2\pi t}} e^{-\frac{1}{2t} \operatorname{arg\,cosh}^2 x} \qquad (t \to 0) \,. \tag{33}$$

By Lerch's theorem [37], if two functions have the same Laplace transform, they can differ at most on a set of measure zero. In this sense, $\hat{\theta}(r,t)$ represents the asymptotic inversion of the Laplace transform (21) in the short-maturity limit.

Remark 2. *Assuming the numerical upper error bound* (17)*, the approximation error of the Laplace transform is bounded uniformly over x as*

$$F_\theta(x,t) = \hat{F}_\theta(x,t)(1+\epsilon(t,x)) \tag{34}$$

where $|\epsilon(t,x)| \leq \frac{1}{70} t$. This follows from

$$|F_\theta(x,t) - \hat{F}_\theta(x,t)| \leq \int_0^\infty e^{-rx} |\theta(r,t) - \hat{\theta}(r,t)| \frac{dr}{r} \leq \frac{1}{70} t \int_0^\infty e^{-rx} \hat{\theta}(r,t) \frac{dr}{r} \,, \tag{35}$$

which is equivalent with the stated bound.

Proof of Proposition 4. Changing the integration variable in (32) to $\rho = rt$, the integral becomes

$$\begin{aligned}\hat{F}_\theta(x,t) &= \int_0^\infty e^{-rx}\hat{\theta}(r,t)\frac{dr}{r} = \frac{1}{2\pi t}\int_0^\infty e^{-\rho(x/t)}G(\rho)e^{-\frac{1}{t}(F(\rho)-\frac{\pi^2}{2})}\frac{d\rho}{\rho} \\ &= \frac{1}{2\pi t}\int_0^\infty G(\rho)e^{-\frac{1}{t}\tilde{F}(\rho,x)}\frac{d\rho}{\rho}\end{aligned} \tag{36}$$

with $\tilde{F}(\rho,x) := F(\rho) + x\rho - \frac{\pi^2}{2}$. Convergence of the integral follows from an examination of the tail asymptotics of the functions F, G as $\rho \to \{0,\infty\}$ given in Propositions 4 and 5 in [32].

We use next the Laplace method to obtain the $t \to 0$ asymptotics of this integral. The functions F, G and their derivatives are continuous and differentiable for all $\rho > 0$. It is shown in the proof of Proposition 4.2 in [29] that F, G are analytical along the real positive axis $\rho > 0$. This justifies the application of the Laplace asymptotic method.

The function $F(\rho)$ is convex on $\rho \geq 0$, such that $\tilde{F}(\rho,x)$ has a unique infimum for any x. Denote this infimum ρ_*, which is given by the solution of the equation

$$F'(\rho_*) + x = 0\,. \tag{37}$$

The $t \to 0$ asymptotics of the integral in the second line of (36) is

$$\frac{1}{2\pi t}\int_0^\infty G(\rho)e^{-\frac{1}{t}\tilde{F}(\rho,x)}\frac{d\rho}{\rho} \sim \frac{1}{\sqrt{2\pi t}}\frac{G(\rho_*)}{\rho_*\sqrt{\tilde{F}''(\rho_*)}}e^{-\frac{1}{t}\tilde{F}(\rho_*)} \tag{38}$$

We will prove that this coincides precisely with $F_\theta(x,t)$ given in (21). The proof has several steps.

Step 1. Compute ρ_*. Recall the main properties of $F(\rho)$ from Section 2 in [32]. This function is convex for all $\rho > 0$ and its derivative takes the value $F'(1) = -1$. From convexity, $F'(\rho) < -1$ holds only for $\rho \in (0,1)$, which implies that the equation (37) with $x \geq 1$ must have a unique solution in $0 < \rho_* < 1$.

For $0 < \rho < 1$, we have $F'(\rho) = -\cosh z$ where z solves the equation $\rho = \frac{z}{\sinh z}$. The equation (37) for ρ_* reads $\cosh z = x$ which gives $z = \arg\cosh x$. The result for ρ_* is

$$\rho_* = \frac{z}{\sinh z} = \frac{\arg\cosh x}{\sqrt{x^2-1}}\,. \tag{39}$$

Step 2. Using this result for ρ_* we compute the exponential factor in (38), which is

$$\begin{aligned}\tilde{F}(\rho_*) &= F(\rho_*) + x\rho_* - \frac{\pi^2}{2} \\ &= \frac{1}{2}z^2 - \rho\cosh z + x\frac{z}{\sinh z} = \frac{1}{2}\arg\cosh^2 x\,.\end{aligned} \tag{40}$$

Step 3. Finally compute the ratio in (38). The functions appearing in this ratio are

$$\tilde{F}''(\rho_*) = F''(\rho_*) = \frac{\sinh^2 z}{\frac{z}{\tanh z} - 1} \tag{41}$$

$$G(\rho_*) = \frac{z}{\sqrt{\frac{z}{\tanh z} - 1}} \tag{42}$$

with $z = \arg\cosh x$. This gives

$$\frac{G(\rho_*)}{\rho_*\sqrt{\tilde{F}''(\rho_*)}} = \frac{z}{\sqrt{\frac{z}{\tanh z} - 1}} \cdot \frac{\sqrt{\frac{z}{\tanh z} - 1}}{\sinh z} \cdot \frac{\sinh z}{z} = 1\,. \tag{43}$$

Substituting the different factors into (38) yields the stated result (33).

□

5. Numerical tests and application

We present in this section several tests and one application of the analytical approximation (18) for the conditional distribution. For these tests we require a good approximation for the functions $F(\rho), G(\rho)$. Although they are exactly known, the exact result requires the solution of an equation for each evaluation, which is inconvenient for practical applications, especially when these functions appears in integrands.

An efficient approximation for these functions was proposed in [29]. They are represented as series expansions in $\log\rho$ within their convergence domain $|\log\rho| < 3.49295$. Outside of this region they are approximated by their tail asymptotics obtained in [32].

Explicitly, we have the approximations

$$F_N(\rho) := \begin{cases} F_L(\rho) & \rho < \rho_L \\ \sum_{k=0}^{N} d_{F,k}\log^k(1/\rho) & \rho \in [\rho_L, \rho_R] \\ F_R(\rho) & \rho > \rho_R \end{cases} \tag{44}$$

and

$$G_N(\rho) := \begin{cases} G_L(\rho) & \rho < \rho_L \\ \sqrt{3}\sum_{k=0}^{N} d_{G,k} \log^k(1/\rho) & \rho \in [\rho_L, \rho_R] \\ G_R(\rho) & \rho > \rho_R \end{cases} \tag{45}$$

The functions $F_{L,R}(\rho), G_{L,R}(\rho)$ denote the asymptotic expansions of the respective functions for $\rho \to \{0, \infty\}$, respectively. They are given in Propositions 4 and 5 of [32]. The first two coefficients $d_{F,k}$ and $d_{G,k}$ are given in Appendix A and the first 10 coefficients are given in Table 2 of [29]. For the numerical evaluations in this section we truncated these series expansions to the first $N = 8$ terms and used $\rho_L = 0.04, \rho_R = 32.88$. Implementations of these functions are available at `https://github.com/dan-pirjol/HartmanWatson`.

5.1. Numerical tests of the conditional density

We present in this section tests for the accuracy of the analytical approximation (18) for the conditional distribution for several values of t.

Tests for the conditional density. Define the normalized asymptotic conditional distribution

$$f^{(0)}_{a_t|s_t}(a, s, t) := \frac{1}{n_c(s, t)} \hat{f}_{a_t|s_t}(a, s, t) \tag{46}$$

where

$$n_c(s, t) := \int_0^\infty \hat{f}_{a_t|s_t}(a, s, t) \frac{da}{a} \tag{47}$$

is a normalization integral.

Figure 1 shows the plots of the normalized conditional density $f^{(0)}_{a_t|s_t}(a, s, t)$ (solid black curve) comparing with the exact result (10) (red dots) obtained by numerical evaluation of the integral for $\theta(r, t)$, for $t = 1.0$ and three values of s as shown.

The dashed lines show the improved log-normal approximation which matches the first two moments of the conditional distribution, given by the exact results (12) and (13). The approximation based on the asymptotic expansion (18) agrees much better with the direct numerical evaluation of (10) than the moment matching approximation.

Tests for the moments of the conditional distribution. For this test we use the analytical approximation (18) to compute the first two moments of

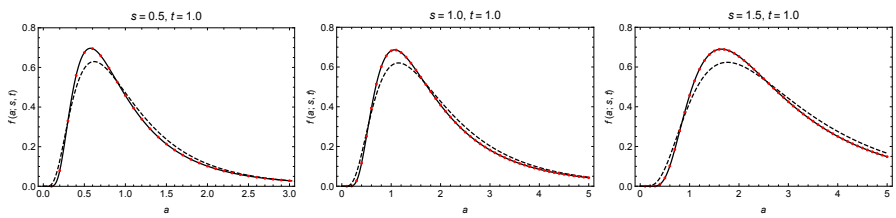

Figure 1: Plots of the normalized asymptotic conditional distribution $f^{(0)}_{a_t|s_t}(a,s,t)$ defined in (46) (solid black) and of the exact result (10) obtained by numerical integration (red dots) for $t = 1.0$ and three values of $s = 0.5, 1.0, 1.5$. The dashed black curves show the log-normal approximation which matches the first two moments of $a_t|s_t$.

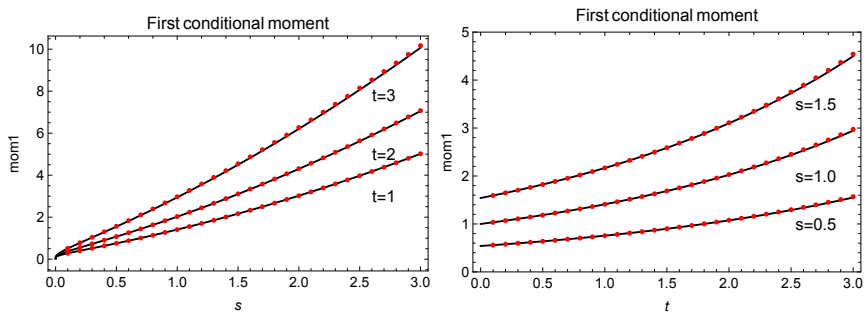

Figure 2: Plots of the first conditional moment of a_t vs s (left) and vs t (right). The solid black curves show the asymptotic result, and the red dots show the exact result (12).

the conditional distribution $\mathbb{E}[a_t|s_t = s]$ and $\mathbb{E}[(a_t)^2|s_t = s]$. Exact results for these moments have been obtained by [19, 28], see (12) and (13).

The moments were computed by evaluating one-dimensional integrals over the normalized density $f^{(0)}_{a_t|s_t}(a,s,t)$ defined in (46). The results are shown in Figure 2 (solid black curves) for several maturities $t = \{1, 2, 3\}$ and values of $s_t = \{0.5, 1, 1.5\}$. The red dots show the exact result (13). We note very good agreement for maturities up to $t \leq 3$.

5.2. Option pricing in the log-normal SABR model

The log-normal SABR model [14] is defined by the process

$$dS_t = \sigma_t S_t dW_t \,, \quad d\sigma_t = \omega \sigma_t dZ_t \tag{48}$$

where $\text{corr}(W_t, Z_t) = \rho$. The volatility process is a driftless geometric Brownian motion. This model is a particular case of the Hull-White model [16] which allows also a drift for the volatility process.

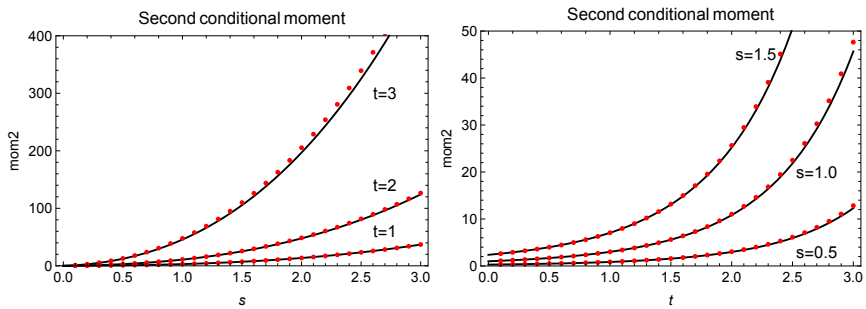

Figure 3: Plots of the second conditional moment of a_t vs s (left) and vs t (right). The solid black curves show the asymptotic result, and the red dots show the exact result (13).

The solution for S_T can be written in closed form by introducing the integrated variance $V_T = \int_0^T \sigma_t^2 dt$. Conditional on a path of the volatility $\{\sigma_t\}_{0\leq t\leq T}$, the asset price S_T is log-normally distributed [38]. We write this as

$$S_T = F(V_T, \sigma_T) e^{\rho_\perp \sqrt{V_T} Z - \frac{1}{2}\rho_\perp^2 V_T}, \quad F(V_T, \sigma_T) = S_0 e^{\frac{\rho}{\omega}(\sigma_T - \sigma_0) - \frac{1}{2}\rho^2 V_T}. \tag{49}$$

with $Z = N(0,1)$ a standard normal random variable and $\rho_\perp = \sqrt{1-\rho^2}$.

This relation is the basis of the mixing formula [22], which gives the option prices as mixtures of Black-Scholes prices

$$C(K,T;S_0) = \int_0^\infty ds \int_0^\infty dv p_T(v,s)[F(v,s)\Phi(d_1) - K\Phi(d_2)] \tag{50}$$

$$P(K,T;S_0) = \int_0^\infty ds \int_0^\infty dv p_T(v,s)[K\Phi(-d_2) - F(v,s)\Phi(-d_1)] \tag{51}$$

where $p_T(v,s)$ is the joint distribution of (V_T, σ_T) defined as

$$\mathbb{P}\Big(V_T \in dv, \sigma_T \in ds\Big) := p(v,s) dv ds \tag{52}$$

and we denoted $F(v,s) = S_0 e^{\frac{\rho}{\omega}(s-\sigma_0) - \frac{1}{2}\rho^2 v}$. Furthermore,

$$d_{1,2} = \frac{1}{\rho_\perp \sqrt{v}} (\log(F(v,s)/K) \pm \frac{1}{2}\rho_\perp^2 v). \tag{53}$$

The option prices can be expressed in terms of the joint distribution $f(a,s,t)$ introduced in Section 2 as follows.

Proposition 5. *The European option prices in the correlated log-normal SABR model* (48) *are*

$$C(K,T) = S_0 \int_0^\infty \frac{da}{a} \int_0^\infty ds f(a,s,\tau) c_{BS}(a,s;K/S_0,\rho) \tag{54}$$

$$P(K,T) = S_0 \int_0^\infty \frac{da}{a} \int_0^\infty ds f(a,s,\tau) p_{BS}(a,s;K/S_0,\rho) \tag{55}$$

where

$$c_{BS}(a,s;k,\rho) := e^{\frac{\rho}{\omega}\sigma_0(s-1)-\frac{1}{2}\rho^2(\sigma_0^2 T)a}\Phi(d_1) - k\Phi(d_2) \tag{56}$$

$$p_{BS}(a,s;k,\rho) := -e^{\frac{\rho}{\omega}\sigma_0(s-1)-\frac{1}{2}\rho^2(\sigma_0^2 T)a}\Phi(-d_1) + k\Phi(-d_2) \tag{57}$$

and

$$d_{1,2}(a,s;k,\rho) = \frac{1}{\rho_\perp\sqrt{(\sigma_0^2 T)a}}\Big[-\log k + \frac{\rho}{\omega}\sigma_0(s-1) - \frac{1}{2}\rho^2(\sigma_0^2 T)a \pm \frac{1}{2}\rho_\perp^2(\sigma_0^2 T)a\Big]. \tag{58}$$

Proof. The joint distribution $p_T(v,s)$ in (50) and (51) can be expressed in terms of the joint density $f(a,s,t)$ for the standardized quantities $A_t^{(\mu)}$ and $s_t^{(\mu)}$ introduced in Section 1.

Using the rescaling property of the standard Brownian motion we have

$$\sigma_T = \sigma_0 e^{\omega Z_T - \frac{1}{2}\omega^2 T} = \sigma_0 e^{Z_{\omega^2 T} - \frac{1}{2}\omega^2 T} = \sigma_0 e^{Z_\tau - \frac{1}{2}\tau} \tag{59}$$

where we introduced the reduced time variable $\tau = \omega^2 T$. In a similar way we have

$$V_T = \sigma_0^2 \int_0^T e^{2(\omega Z_s - \frac{1}{2}\omega^2 s)} ds = \frac{\sigma_0^2}{\omega^2}\int_0^{\omega^2 T} e^{2(Z_u - \frac{1}{2}u)} du = \frac{\sigma_0^2}{\omega^2} A_{\omega^2 T}^{(-1/2)}. \tag{60}$$

In conclusion, the variables V_T, σ_T are expressed in terms of the standardized quantities

$$A_t^{(\mu)} = \int_0^t e^{2(B_s + \mu s)} ds, \quad s_t^{(\mu)} = e^{Z_t + \mu t}. \tag{61}$$

as

$$V_T = \frac{\sigma_0^2}{\omega^2} A_{\omega^2 T}^{(-1/2)}, \quad s_T = \sigma_0 s_{\omega^2 T}^{(-1/2)}. \tag{62}$$

The joint density $p_T(v,s)$ is thus expressed as

$$p_T(v,s) = \frac{1}{\sigma_0^3 t} f(v/(\sigma_0^2 T), s/\sigma_0, \omega^2 T) \tag{63}$$

Substituting into (50) and (51) we get the representations of the option prices as double integrals in Proposition 5. □

A similar approach can be used to obtain the joint density of the asset price and volatility in the SABR model by expressing it in terms of the joint distribution $f(a,s,t)$. This is the basis of the approach of Wu (2012) [39] which derived an expansion for the joint density in the total vol-of-vol parameter $\omega^2 t$ and used it to obtain an expansion for the option prices.

Uncorrelated limit. In the uncorrelated limit $\rho = 0$, the functions $c_{BS}(a,s;k,\rho)$ and $p_{BS}(a,s;k,\rho)$ depend only on a, but not on s. Thus the option prices depend only on the density of the time-average of the geometric Brownian motion

$$f_A(a,\tau) := \int_0^\infty f(a,s,\tau) ds\,. \tag{64}$$

The option prices are given by one-dimensional integrals

$$C(K,T;\rho=0) = S_0 \int_0^\infty \frac{da}{a} f_A(a,\tau) c_{BS}^{(\rho=0)}(a; K/S_0, 0) \tag{65}$$

and similar for the put, with $c_{BS}^{(\rho=0)}(a;k,0) := \Phi(d_1) - k\Phi(d_2)$ with $d_{1,2} = \frac{1}{\sqrt{(\sigma_0^2 T)a}}\Big(-\log k \pm \frac{1}{2}(\sigma_0^2 T)a\Big)$.

5.2.1. Master formula for option pricing

The double integrals in (54) and (55) are in fact triple integrals if the Hartman-Watson function $\theta(r,t)$ has to be evaluated using the integral representation (8). This is inconvenient for applications. We propose here the use of the asymptotic expansion of the Hartman-Watson function in the regime $t \to 0$ at fixed rt discussed in Section 3. This approach reduces the pricing problem to the evaluation of a two-dimensional integral.

We propose to replace $f(a,s,t)$ with its analytical approximation described above in Section 3

$$f(a,s,t) \to \frac{1}{n(t,\mu)} f_0(a,s,t)\,, \quad f_0(a,s,t) := f_{LN}(s,t;\mu)\hat{f}_{a_t|s_t}(a,s,t) \tag{66}$$

with $\hat{f}_{a_t|s_t}(a,s;t)$ given in Proposition 1 and $n(t,\mu)$ is a normalization factor given by

$$n(t,\mu) = \int_0^\infty f_0(a,s,t)\frac{dads}{a}\,. \tag{67}$$

Note that this normalization factor is different from the normalization factor of the conditional distribution $n_c(s,t)$ defined above in (47), and is related to it as $n(t,\mu) = \int_0^\infty n_c(s,t) f_{LN}(s,t;\mu)ds$. Working with this normalization factor simplifies the option pricing formula by avoiding integration over $n_c(s,t)$. We also note that $n_c(s,t)$ is independent of the drift parameter μ, in contrast to $n(t,\mu)$ which depends on this parameter.

The normalization factor $n(t,\mu)$ can be reduced to a one-dimensional integral, by making the change of variable $s = \lambda a$ and performing the a integration in closed form. This gives

$$\begin{aligned} n(t,\mu) &= \int_0^\infty d\lambda \int_0^\infty da f_0(a,\lambda a,t) \qquad (68)\\ &= \frac{1}{2\pi t} e^{-\frac{1}{2}\mu^2 t} \int_0^\infty d\lambda \lambda^{\mu-1} G(\lambda) e^{-\frac{1}{t}(F(\lambda)-\pi^2/2)} \int_0^\infty da a^{\mu-1} e^{-\frac{1+\lambda^2 a^2}{2at}} \\ &= \frac{1}{\pi t} e^{-\frac{1}{2}\mu^2 t} \int_0^\infty G(\lambda) K_{-\mu}(\lambda/t) e^{-\frac{1}{t}(F(\lambda)-\pi^2/2)} d\lambda \end{aligned}$$

where $K_n(x)$ is the modified Bessel function of order n.

Substituting the approximation (66) for the conditional density into the option pricing formulas (54), (55) gives a two-dimensional integral representation for option prices in the SABR model, which we call the master formula.

Corollary 1 (Master formula)**.** *The option prices in the log-normal SABR model are approximated by the two-dimensional integrals*

$$\bar{C}(K,T) = S_0 \frac{1}{n(\tau,-1/2)} \int_0^\infty da \int_0^\infty d\lambda f_0(a,\lambda a,\tau) c_{BS}(a,a\lambda;k,\rho) \tag{69}$$

$$\bar{P}(K,T) = S_0 \frac{1}{n(\tau,-1/2)} \int_0^\infty da \int_0^\infty d\lambda f_0(a,\lambda a,\tau) p_{BS}(a,a\lambda;k,\rho)\,. \tag{70}$$

with $\tau = \omega^2 T$ and $f_0(a,s,\tau) = f_{LN}(s,\tau;-\frac{1}{2})\hat{f}_{a_t|s_t}(a,s,\tau)$ with $f_{LN}(s,\tau;\mu)$ given in (4)*. The approximation error of this result is $C(K,T) = \bar{C}(K,T)(1+O(\omega^2 T))$ and analogous for $P(K,T)$.*

The drift parameter has been taken equal to $\mu = -1/2$, see (62). This is related to the driftless process for σ_t assumed in the SABR model (48).

5.2.2. Numerical tests

We used the master formulas (69), (70) to price ATM options in the log-normal SABR model with several maturities. We assume the following parameters

$$\sigma_0 = 0.2\,, \quad \omega = 1.0\,, \quad S_0 = 1000\,. \tag{71}$$

Option prices are obtained by evaluating the 2-dimensional integrals in the master formulas (69) and (70).

We compare the prices of call options obtained from the master formula 69 with benchmark evaluations obtained using the (exact) transform method, see [23] and Sec. 8.16 in [24]. Table 1 shows the test results for the uncorrelated case $\rho = 0$, and Table 2 shows the test results for negative correlation $\rho = -0.75$. The benchmark values for this case are the same as in Table 8.6 in Lewis (2016) [24]. For both cases we note good agreement with the benchmarks for short maturities, up to about 1 year. For longer maturities the differences increase but remain below 5% for all maturities up to 5 years. In Figure 4 we compare the computed implied volatility with the benchmarks from Tables 1 and 2. The plots show good agreement.

Tables 1 and 2 show also the evaluation times for the method proposed here and for the benchmark method. The run times (in seconds) shown are obtained in Mathematica 14.1.0 running on an iMac with Mac OS X ARM (64-bit) Sequoia 15.7.4.

The method proposed in this paper outperforms in speed the benchmark approach [24] for shorter maturities ($T = 0.25$). At larger maturities, the speeds are comparable, and in certain cases the benchmark method is faster. In practical applications, some additional speed-up can be achieved at the expense of accuracy, for example by truncating the expansions for F, G to a smaller number of terms ($N = 8$ was used for the examples shown).

Remark 3. *We comment on the option pricing method in the log-normal SABR model which was used for benchmarking. This is the transform method presented in Sec. 8.16 of Lewis vol.2 [24] which represents option prices as an integral over the characteristic function, which is expressed in closed form as a 1-dimensional integral. This reduces the problem to the evaluation of a double integral. An alternative exact option pricing method was given by Antonov et al [1] in their equation (3.103) (p. 73). This requires the evaluation of a double integral with oscillatory factors which make the numerical evaluation more challenging (especially for small maturity).*

Table 1: Numerical tests for European options in the uncorrelated SABR model $\rho = 0$ and parameters $\sigma_0 = 0.2, \omega = 1$ for several strikes and maturities, from the master formula (69). The benchmark valuations are obtained using the transform method of [23]. The run time τ [sec] is compared with that for the benchmark evaluation τ_b [sec].

	$T = 0.25$				$T = 1.0$			
K	$\bar{C}(K,T)$	benchmark	τ	τ_b	$\bar{C}(K,T)$	benchmark	τ	τ_b
500	500.001	500.001	5.4	18.5	502.516	502.517	5.1	5.6
750	250.406	250.406	5.4	18.0	264.855	264.859	5.4	6.4
1000	40.6875	40.6876	5.5	17.9	85.4432	85.449	5.6	6.4
1250	1.4715	1.4716	5.0	17.9	26.4868	26.4928	5.2	6.4
1500	0.0976	0.0976	3.9	17.9	12.3927	12.3967	4.5	5.7
1750	0.0132	0.0132	3.5	18.0	7.3826	7.3859	4.0	5.7
2000	0.00283	0.00283	3.2	18.0	5.03172	5.03492	3.8	5.7
	$T = 2.0$				$T = 5.0$			
K	$\bar{C}(K,T)$	benchmark	τ	τ_b	$\bar{C}(K,T)$	benchmark	τ	τ_b
500	512.660	512.688	5.1	4.4	535.417	536.211	6.1	3.9
750	289.194	289.244	5.4	4.4	330.946	331.211	6.3	3.9
1000	124.242	124.308	5.2	4.5	180.527	180.858	5.9	3.9
1250	61.3959	61.461	5.1	4.5	118.179	118.526	5.3	3.9
1500	40.5649	40.626	4.5	4.5	93.8559	94.208	5.1	3.9
1750	30.8797	30.938	4.3	4.6	80.9812	81.327	4.7	3.9
2000	25.3223	25.938	4.0	4.5	72.8452	73.112	4.5	3.9

5.3. Simulation and extension to other models

The analytical approximation for the conditional distribution presented in Section 3 can be used for the simulation of the log-normal SABR model. The approach is analogous to that used in Cai et al (2017) [7]. The simulation of S_T proceeds in two steps:

i) Simulate $V_T|\sigma_T$. This is done by drawing a sample from the conditional distribution $f_{a_t|s_t}(a, s, t)$. This can be done using the approximation given in (46).

ii) S_T is simulated as a log-normal random variable, conditional on the (V_T, σ_T) obtained in the first step.

The authors of [7] computed an approximation for the joint distribution by inverting the Laplace transform of $f_{a_t|s_t}(a, s, t)$. The analytical approximation discussed here avoids the inversion of a Laplace transform.

The results of this paper can be extended to other volatility models where

Table 2: Numerical tests for European options in the correlated SABR model with negative correlation $\rho = -0.75$ and parameters (71). The column $\bar{C}(K,T)$ shows the result from the master formula (69). The benchmark evaluations are from Table 8.6 in [24]. The run time τ [sec] is compared with that for the benchmark evaluation τ_b [sec].

	$T = 0.25$				$T = 1.0$			
K	$\bar{C}(K,T)$	benchmark	τ	τ_b	$\bar{C}(K,T)$	benchmark	τ	τ_b
500	500.027	500.027	5.8	30.4	505.112	505.114	5.5	7.9
750	251.426	251.426	6.0	24.5	270.715	270.719	5.6	7.0
1000	39.6175	39.6175	5.7	24.6	77.0641	77.069	5.6	7.7
1250	0.03327	0.03327	5.0	24.6	4.8766	4.8787	4.5	7.7
1500	$1.80 \cdot 10^{-5}$	$1.80 \cdot 10^{-5}$	3.2	24.7	0.4909	0.4915	4.0	7.7
1750	$1.45 \cdot 10^{-7}$	$1.47 \cdot 10^{-7}$	3.0	24.8	0.1155	0.1157	3.4	6.9
2000	$4.81 \cdot 10^{-9}$	$4.81 \cdot 10^{-9}$	4.5	24.6	0.04093	0.04103	3.1	7.0
	$T = 2.0$				$T = 5.0$			
K	$\bar{C}(K,T)$	benchmark	τ	τ_b	$\bar{C}(K,T)$	benchmark	τ	τ_b
500	516.178	516.207	5.0	6.2	535.012	535.141	5.7	5.4
750	291.981	292.024	5.3	5.5	320.845	321.021	5.6	4.4
1000	103.323	103.370	5.5	5.5	135.668	135.859	5.7	4.4
1250	15.5282	15.557	4.5	5.5	34.8173	34.951	4.8	4.4
1500	3.6624	3.6761	4.2	5.5	13.0739	13.159	4.6	4.4
1750	1.4596	1.4674	3.8	5.5	6.92228	6.978	4.2	4.4
2000	0.7463	0.7513	3.5	5.5	4.3134	4.354	4.0	4.4

the volatility follows a driftless geometric Brownian motion. As discussed in [19], these models include

$$\text{Normal SABR: } dS_t = \sigma_t dW_t \tag{72}$$

$$\text{Log-normal SABR: } dS_t = \sigma_t S_t dW_t \tag{73}$$

$$\text{CEV-SABR: } dS_t = \sigma_t S_t^{\beta} dW_t\,, \quad \beta \in (0,1) \tag{74}$$

$$\text{Displaced Diffusion SABR: } dS_t = (S_t + \theta)\sigma_t dW_t \tag{75}$$

where σ_t follows $d\sigma_t = \omega\sigma_t dZ_t$, with (W_t, Z_t) correlated standard Brownian motions with correlation ρ.

We comment also on option pricing in the general Hull-White model [16]. This corresponds to adding a drift to the volatility process $d\sigma_t = \omega\sigma_t dZ_t + \mu_\sigma \sigma_t dt$. Conditional on a realization of the volatility process $\{\sigma_t\}_{0\le t\le T}$, the asset price S_T is log-normally distributed and can be expressed in terms of

$V_T = \int_0^T \sigma_t^2 dt, U_T = \int_0^T \sigma_t dt$ and σ_T [21]. The result (49) is straightforwardly extended to this case as

$$S_T = F(V_T, U_T, \sigma_T) e^{\rho_\perp \sqrt{V_T} Z - \frac{1}{2}\rho_\perp^2 V_T}, F(V_T, U_T, \sigma_T) = S_0 e^{\frac{\rho}{\omega}(\sigma_T - \sigma_0) - \frac{\rho \mu_\sigma}{\omega} U_T - \frac{1}{2}\rho^2 V_T}. \tag{76}$$

The option pricing problem requires the trivariate joint distribution of $(\sigma_\tau, \int_0^\tau \sigma_s ds, \int_0^\tau \sigma_s^2 ds)$ [21], which goes beyond the approach used here. We note that Alili, Matsumoto and Shiraishi (2004) [2] studied this trivariate distribution and gave expressions for its Laplace transform.

In the uncorrelated case $\rho = 0$ the result simplifies greatly, as only the distribution of V_T is required[1]. The Master formula for option pricing in Corollary 1 can be extended to the uncorrelated Hull-White model with

[1]This case was studied by Forde (2010) [11] under the additional assumption $\mu_\sigma = \frac{1}{2}\omega^2$ (which corresponds to the so-called modified SABR model discussed in Sec. 2.4 of [18]) for which the density of V_T can be found from the Bougerol's identity [26], see also an explicit result for this density as a one-dimensional integral in [9].

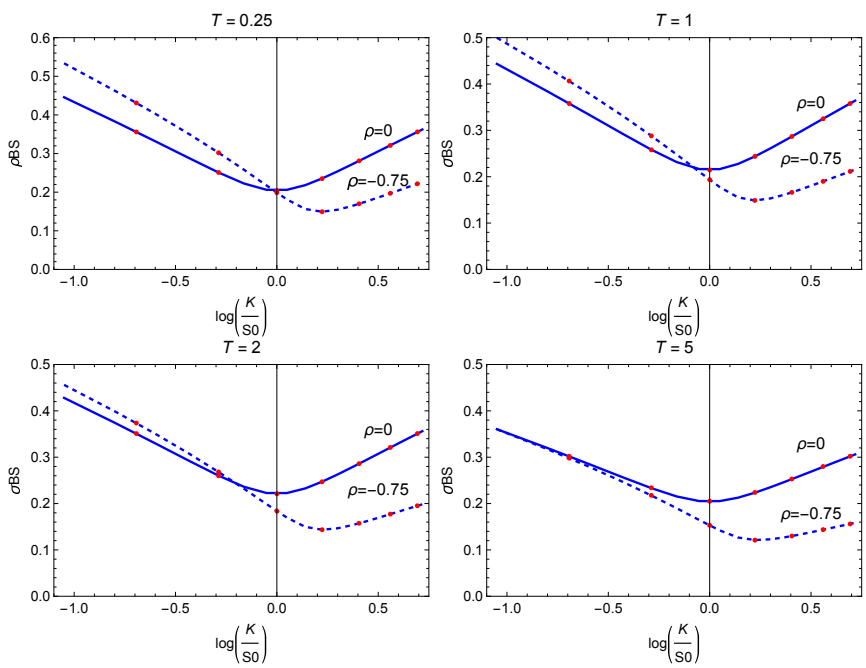

Figure 4: Plots of the implied volatility computed using the method of this paper (blue solid and dashed curves) comparing with the benchmarks from [24] (red dots) for several maturities.

volatility drift μ_σ by taking $\mu = \frac{\mu_\sigma}{\omega^2} - \frac{1}{2}$ in the normalization factor $n(\tau, \mu)$ and $f_{LN}(s, \tau; \mu)$, and replacing $c_{BS}(a, a\lambda; k, \rho)$ with $c_{BS}^{(\rho=0)}(a; k, 0)$ given in text below equation (65). A similar result holds for the put option.

A similar limitation of the option pricing method discussed here holds also for the CEV-SABR model with $\beta \neq 1$. In this model, an exact representation for the option prices can be written only for the uncorrelated case $\rho = 0$. For this case one can show that the stock price process is equivalent to a time-changed squared Bessel process [18, 7], which leads to an exact option pricing result. The non-zero correlation case can be handled in an approximate way by a mimicking approach [1].

The limitations mentioned for the CEV-SABR model can be avoided by a direct expansion of the pricing PDE for option prices in the total vol-of-vol $\omega\sqrt{T}$. This approach was used by Yang et al (2017) [40] and Grischenko, Han and Nistor (2020) [13] to derive expansions for option prices in this model. We mention also the heat kernel expansion which was used to obtain the first three terms in the short-maturity expansion for the mean-reverting SABR model by Ben-Arous and Laurence (2018) [4].

The analytical approximation proposed here is also relevant for pricing options which are sensitive to the joint distribution of the terminal value and the time integral of a geometric Brownian motion. One particular case of interest are generalized Asian options with payoff $(\theta(xA_T + yS_T - K))^+$ where $\theta = \pm 1$ corresponding to calls/puts, under the Black-Scholes model. This payoff appears naturally when pricing forward starting Asian options during their averaging period. The short maturity asymptotics of these options was obtained in [31] and is expressed in terms of a function which is related to the short-maturity asymptotics of the Hartman-Watson integral.

Appendix A. The functions $F(\rho), G(\rho)$

For convenience we collect in this Appendix the definitions of several functions appearing in the asymptotic expansions discussed in text.

Definition 1. *The function $F(\rho) : \mathbb{R}_+ \to \mathbb{R}_+$ is defined by*

$$F(\rho) = \begin{cases} \frac{1}{2}x^2 - \frac{x}{\tanh x} + \frac{\pi^2}{2} & , 0 < \rho < 1 \\ \frac{1}{2}u^2 - \frac{u}{\tan u} + \frac{1}{2}\pi^2 & , \rho > 1 \\ \frac{\pi^2}{2} - 1 & , \rho = 1 \end{cases} \tag{A.1}$$

and the function $G(\rho) : \mathbb{R}_+ \to \mathbb{R}_+$ is defined by

$$G(\rho) = \begin{cases} \frac{x}{\sqrt{\frac{x}{\tanh x} - 1}} & , 0 < \rho < 1 \\ \frac{u}{\sqrt{1 - \frac{u}{\tan u}}} & , \rho > 1 \\ \sqrt{3} & , \rho = 1 \end{cases} \tag{A.2}$$

In these expressions $x \geq 0$ is the solution of the equation

$$\rho \frac{\sinh x}{x} = 1 \tag{A.3}$$

and $u \in (0, \pi)$ is the solution of the equation

$$\rho \frac{\sin u}{u} = 1 \,. \tag{A.4}$$

These functions can be expanded in power series of $\log(1/\rho)$, which are convenient for numerical evaluation

$$F(\rho) = \frac{\pi^2}{2} - 1 + \log(1/\rho) + \log^2(1/\rho) + O(\log^3(1/\rho)) \tag{A.5}$$

$$G(\rho) = \sqrt{3}\Big(1 + \frac{1}{5}\log(1/\rho) - \frac{1}{70}\log^2(1/\rho) + O(\log^3(1/\rho)\Big) \,. \tag{A.6}$$

The first 10 coefficients in these expansions are given in Table 2 of [29]. Propositions 4.1 and 4.2 in [29] show that these series expansions converge for $|\log(1/\rho)| < 3.49295$. Outside of this range, analytical tail asymptotics for $\rho \to 0$ and $\rho \to \infty$ are available [32] and should be used for a precise approximation.

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References

[1] A. Antonov, M. Konikov and M. Spector, Modern SABR Analytics. Springer, New York 2010

[2] L. Alili, H. Matsumoto and T. Shiraishi, On a triplet of exponential Brownian functionals, Séminaire de Probabilités XXXV, Springer, 2004.

[3] P. Barrieu, A. Rouault and M. Yor, A study of the Hartman-Watson distribution motivated by numerical problems related to the pricing of Asian options, Journal of Applied Probability 41(4), 1049-1058, 2004.

[4] G. Ben Arous and P. Laurence, Second order expansion for implied volatility in two factor local stochastic volatility models and applications to the dynamic λ-SABR model, in *Large Deviations and Asymptotic Methods in Finance*, vol 110, p.89-136, Editors P.K. Friz, J. Gatheral, A. Gulisashvili, A. Jacquier and J. Teichmann, Springer Proceedings in Mathematics and Statistics, 2015.

[5] G. Bernhart and J.F. Mai, A note on the numerical evaluation of the Hartman-Watson density and distribution function. In: K. Blau, M. Scherer and R. Zagst (Eds) *Innovations in quantitative risk management*. Springer Proceedings in Mathematics and Statistics, Springer, p 337, 2015

[6] P. Boyle and A. Potapchik, Prices and sensitivities of Asian options: A survey. *Insurance: Mathematics and Economics* **42**, 189-211 (2008)

[7] N. Cai, Y. Song and N. Chen, Exact simulation of the SABR model, *Operations Research* **65**(4), 931-951 (2017)

[8] J. Choi, L. Hu and Y.K. Kwok, Efficient simulation of the SABR model, arXiv:2408.01898[q-fin]

[9] D. Dufresne, The integral of geometric Brownian motion, Adv. Appl. Prob. 33, 223-241 (2001)

[10] W. Feller, *An introduction to probability theory and its applications*, vol. 2, 2nd edition, Wiley, New York, 1968.

[11] M. Forde, Exact pricing and large-time asymptotics for the modified SABR model and the Brownian exponential functional, 2010

[12] S. Gerhold, The Hartman-Watson distribution revisited: asymptotics for pricing Asian options. J Appl Probab 48(3):892–899 (2011)

[13] O. Grishchenko, X. Han and V. Nistor, A volatility-of-volatility expansion of the option prices in the SABR stochastic volatility model, International Journal of Theoretical and Applied Finance 23(03), 2050018 (2020)

[14] P.S. Hagan, D. Kumar, A.S. Lesniewski and D.E. Woodward, Managing smile risk, Wilmott Magazine Sept. 2002.

[15] P. Hartman and G.S. Watson, "Normal" distribution functions on spheres and the modified Bessel functions, *Ann. Probab.* **2**, 593-607 (1974)

[16] J. Hull and A. White, The pricing of options on assets with stochastic volatilities, Journal of Finance, 17(2), 281-300 (1987).

[17] K. Ishyiama, Methods for evaluating density functions of exponential functionals represented as integrals of geometric Brownian motion. Methodology and Computing in Applied Probability 7:271–283

[18] O. Islah, Solving SABR in exact form and unifying it with LIBOR market model, 2009, `https://ssrn.com/abstract=1489428`

[19] J. Kennedy, S. Mitra and D. Pham, On the approximation of the SABR model: A probabilistic approach, Applied Mathematical Finance 2012.

[20] I. Kyriakou, R. Brignone and G. Fusai, Unified moment-based modeling of integrated stochastic processes, Operations Research 72(4), 1630-1653, 2024.

[21] B. Leblanc, Une approche unifiée pour une forme exacte du prix d'une option dans les différents modèles à volatilité stochastique, *Stochastic and Stochastic Reports*, 57 (1996), 1-35.

[22] A.L. Lewis, The mixing approach to stochastic volatility and jump models, Wilmott magazine, March 2002.

[23] A.L. Lewis, Option valuation under stochastic volatility I. Finance Press, Newport Beach, 2000.

[24] A.L. Lewis, Option valuation under stochastic volatility II. Finance Press, Newport Beach, 2016.

[25] A. Lyasoff, Another look at the integral of the exponential Brownian motion and the pricing of Asian options, Finance and Stochastics 20, 1061-1096, 2016

[26] H. Matsumoto and M. Yor, Exponential functionals of Brownian motion, I: Probability laws at fixed time, Probability Surveys 2, 312-347, 2005.

[27] W.A. McGhee. "An Efficient Implementation of Stochastic Volatility by the Method of Conditional Integration". ICBI Global Derivatives and Risk Management, April 2011.

[28] W.A. McGhee. "Artificial neural network representation of the SABR stochastic volatility model". SSRN 3288882, 2018

[29] P. Nándori and D. Pirjol, On the distribution of the time-integral of the geometric Brownian motion, Journal of Computational and Applied Mathematics 402, 113818, 2022.

[30] D. Pirjol and L. Zhu, Asymptotics of the time-discretized log-normal SABR model: The implied volatility surface, Probability in the Engineering and Informational Sciences 35(4), 942-974, 2021

[31] D. Pirjol, J. Wang and L. Zhu, Short maturity forward start Asian options in local volatility models, Applied Mathematical Finance 26(3), 187-221, 2019

[32] D. Pirjol, Small-t asymptotic expansion of the Hartman-Watson distribution, Methodology and Computing in Applied Probability 23(4), 1537-1549, 2021.

[33] D. Pirjol, Error bound for the asymptotic expansion of the Hartman-Watson integral, arXiv:250.04992[math-CA]

[34] D. Pirjol and L. Zhu, Short maturity Asian options in local volatility models, SIAM Journal on Financial Mathematics 7(1), 947-992, 2016

[35] D. Revuz and M. Yor, *Continuous martingales and Brownian motion*, (Grundlehren der mathematischen Wissenschaften, 293), New York, Springer, 3rd edition, 1998

[36] A. Leitao Rodriguez, L.A. Grzelak and C.W. Oosterlee, On a one time-step SABR simulation approach: Application to European options, Applied Mathematics and Computation 293, 461-479, 2017

[37] D. Widder, *The Laplace transform*, Princeton University Press, Princeton, 1946.

[38] G. Willard, Calculating prices and sensitivities for path-independent derivative securities in multifactor models, Journal of Derivatives 5 45–61 (1997)

[39] Q. Wu, Series expansion of the SABR joint density. Mathematical Finance 22(2):310–345, 2012.

[40] N. Yang, N. Chen, N. Liu and X. Wan, Approximate arbitrage-free option pricing under the SABR model, Journal of Economic Dynamics and Control 83, 198-214, 2017

[41] M. Yor, On some exponential functionals of Brownian motion, Advances in Applied Probability 24, 509-531, 1992